Chapter 5

Empirical Data Analysis

This chapter applies the proposed FARFIMA order-selection and forecasting framework to two French demographic applications. The first considers age-specific fertility-rate changes, while the second considers female and male log-mortality rates. Both datasets have a natural functional structure because each annual observation is a curve indexed by age. They nevertheless represent different forms of persistence. Fertility-rate changes are compatible with stationary long memory, whereas mortality is close to first-order integration. This contrast is useful for identifying two distinct sources of forecast improvement: estimating the fractional order d in the fertility application and selecting the short-memory FARMA orders (p, q) in the mortality application.

The empirical analysis follows the functional demographic workflow of smoothing noisy age-specific rates before modelling them as curves (Hyndman et al., 2007; Hyndman et al., 2009), with rainbow plots used to display their temporal evolution as in Shang et al. (2025). For each application, the functional autocorrelation function (fACF) of Mestre et al. (2021) and the stationarity test of Horváth et al. (2014) assess persistence, while the bias-corrected LPWN estimator determines the fractional order. The three forecasting methods are then evaluated using expanding-window forecasts and the point and interval criteria introduced in Chapter 3.

5.1 Forecasting Design

The forecasting experiment compares three methods. The benchmark is a FARFIMA$(1, 1, 0)$ model with $d = 1$. Functional IC estimates d and selects (p, q) using the proposed adaptive functional information criterion. CV uses the same estimate of d but selects (p, q) by expanding-window cross-validation. The selected orders are fixed before evaluation on the test sample.

For each series, the first 70% of curves form the training sample, the next 15% form the validation sample, and the final 15% form the test sample. Forecasts are produced using an expanding estimation window for horizons $h = 1, \ldots, 10$. Candidate orders satisfy $0 \leq p, q \leq 5$, excluding $(0, 0)$. The memory estimator uses one principal component, the empirical covariance operator, 399 bootstrap replications, and a fractional-differencing trim of 20. Central 80% and 95% split conformal prediction intervals are calibrated on the validation sample.

Models and conformal bands are estimated from the smoothed curves, but all reported metrics are calculated against the corresponding raw observations. The evaluation therefore includes both forecast error in the underlying smooth curve and observation-level variation. This is more demanding than evaluation against smoothed curves, particularly for mortality at short horizons and for fertility near the boundary ages.

5.2 Empirical Studies

5.2.1 French Fertility-Rate Changes

Data source and preprocessing

The fertility data are annual age-specific fertility rates for France obtained from the Human Fertility Database (Human Fertility Database, 2026). The dataset and its country documentation are available from the Human Fertility Database France page. The archived sample used in this study covers 1946–2022 and contains one curve for each year over ages 15–49.

Let $F_t(u)$ denote the fertility rate at age u in year t. The analysis focuses on the symmetric relative change

$$C_t(u) = \frac{2\{F_{t-1}(u) - F_t(u)\}}{F_{t-1}(u) + F_t(u)}, \qquad t = 1947, \ldots, 2022. \tag{5.1}$$

Thus, $C_t(u) > 0$ indicates a decline in the age-specific fertility rate from year $t-1$ to year t, whereas $C_t(u) < 0$ indicates an increase. The transformation reduces the influence of the overall scale and produces 76 annual change curves.

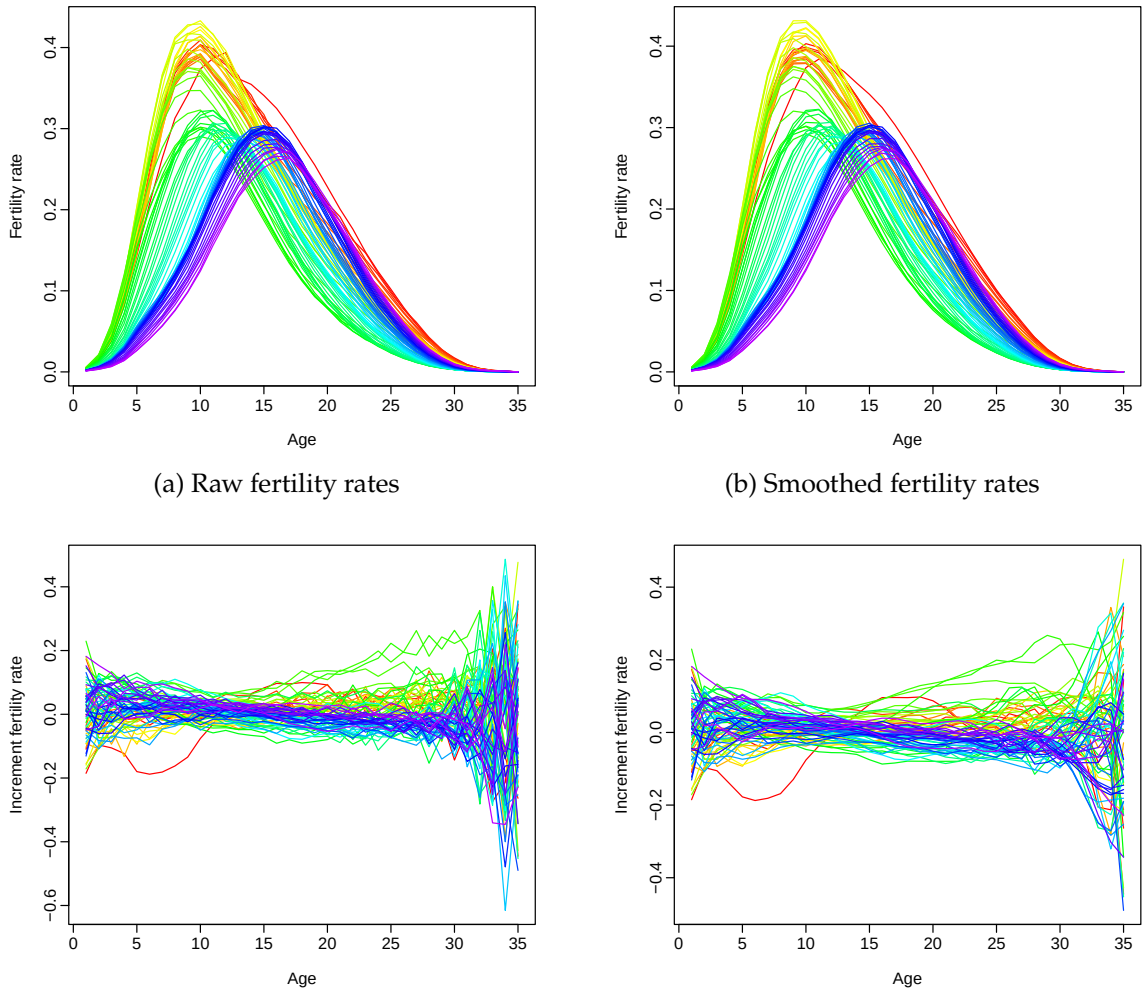
The archived smoothed fertility curves are used for model estimation. The smoothing follows the functional demographic approach of estimating a smooth age profile in each year, with population-based weights accounting for heterogeneous measurement variability (Hyndman et al., 2007). Equation (5.1) is calculated separately from the raw and smoothed rates. Models are fitted to the smoothed changes, while forecast accuracy is evaluated against the raw changes.

Figure 5.1 compares the raw and smoothed fertility rates and their corresponding changes. Both versions show a clear shift of the modal childbearing age towards older ages and a reduction in the peak fertility rate. Smoothing removes local age-specific variation while retaining this evolution. The change curves fluctuate around zero across most ages but are more variable near the boundaries, where the fertility rates in the denominator of (5.1) are small.

Persistence and selected orders

The stationarity null is not rejected ($p = 0.177$), while the estimated memory parameter is $\widehat{d} = 0.3886$. This estimate lies in the stationary long-memory range $0 < d < 1/2$. Figure 5.2 provides complementary evidence: the lag-one fACF is 0.247, the value at lag five is 0.160, and lags 1–5 and several later lags exceed the estimated 95% white-noise bound of 0.146. The fACF therefore decays persistently rather than displaying an abrupt short-memory cut-off.

Functional IC selects $(p, q) = (1, 0)$ and CV selects $(1, 1)$. Their short-memory orders are similar, and both use the same estimate $\widehat{d} = 0.3886$. The principal difference from the



(a) Raw fertility rates

(b) Smoothed fertility rates

(c) Raw increment fertility rate

(d) Smoothed increment fertility rate

Figure 5.1: Rainbow plots of the raw and smoothed French age-specific fertility rates and their increments. Curves run chronologically from the earliest years in red to the most recent years in violet.

FARFIMA$(1, 1, 0)$ benchmark is therefore the estimated fractional order rather than the selected FARMA order.

Forecasting results

Tables 5.1 and 5.2 show that both data-driven methods outperform the benchmark across the ten horizons. At $h = 10$, Functional IC reduces RMSE from 0.1034 to 0.0665, MAE from 0.0626 to 0.0426, and WIS from 0.0278 to 0.0193. CV gives almost identical results. Because the two selected short-memory orders differ only by one MA term but both replace $d = 1$ by $\widehat{d} = 0.3886$, the pattern indicates that the main gain comes from estimating the memory parameter correctly. This interpretation is consistent with the stationary long-memory diagnostics, although a formal attribution would require an additional ablation experiment holding (p, q) fixed.

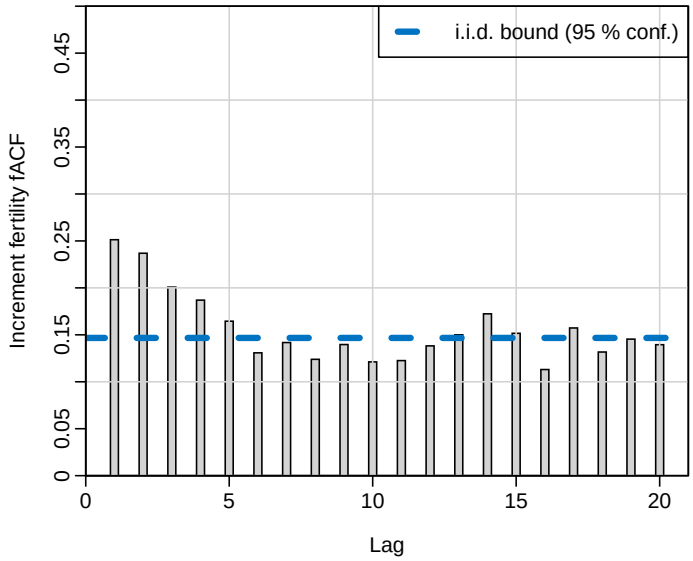


Figure 5.2: Functional autocorrelation function of the French fertility-rate changes. The dashed blue line is the simulated upper 95% white-noise bound.

Table 5.1: Point forecast accuracy for French fertility-rate changes over all ten horizons. The smallest RMSE is red and the smallest MAE is blue.

	RMSE			MAE		
h	Bench.	IC	CV	Bench.	IC	CV
1	0.0744	0.0586	**0.0583**	0.0425	0.0341	**0.0338**
2	0.0672	0.0587	**0.0584**	0.0391	0.0332	**0.0329**
3	0.0726	0.0564	**0.0563**	0.0430	**0.0339**	**0.0339**
4	0.0714	0.0571	**0.0569**	0.0421	0.0348	**0.0347**
5	0.0720	**0.0565**	**0.0565**	0.0438	0.0360	**0.0358**
6	0.0812	**0.0594**	**0.0594**	0.0500	**0.0377**	0.0378
7	0.0855	**0.0599**	0.0600	0.0525	**0.0383**	**0.0383**
8	0.0794	**0.0599**	0.0600	0.0503	**0.0393**	0.0394
9	0.0937	0.0643	**0.0642**	0.0578	**0.0422**	**0.0422**
10	0.1034	0.0665	**0.0664**	0.0626	**0.0426**	**0.0426**

The 95% intervals from Functional IC and CV attain high coverage and lower interval scores than the benchmark. Their 80% intervals tend to over-cover: the mean CPD is approximately 0.11, compared with 0.072 for the benchmark. The interval scores and WIS should therefore be interpreted jointly with coverage rather than in isolation.

Table 5.2: Interval forecast performance for French fertility-rate changes over all ten horizons. The smallest CPD is red, the smallest interval score is blue, and the smallest WIS is purple. WIS combines both coverage levels and is reported once per horizon in the 95% row.

		ECP			CPD			Score			WIS		
h	CP	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV
1	95%	0.9179	0.9750	0.9589	0.0321	0.0250	**0.0089**	0.2789	0.2075	**0.2033**	0.0196	0.0152	**0.0151**
	80%	0.8232	0.8429	0.8554	**0.0232**	0.0429	0.0554	0.2065	0.1586	**0.1583**			
2	95%	0.9886	0.9905	0.9905	**0.0386**	0.0405	0.0405	0.2500	**0.2118**	0.2124	0.0177	**0.0149**	**0.0149**
	80%	0.9067	0.9086	0.9124	**0.1067**	0.1086	0.1124	0.1849	**0.1549**	0.1556			
3	95%	0.9837	0.9939	0.9959	**0.0337**	0.0439	0.0459	0.2451	**0.2246**	0.2250	0.0187	0.0154	**0.0153**
	80%	0.9020	0.9367	0.9306	**0.1020**	0.1367	0.1306	0.1921	0.1580	**0.1565**			
4	95%	0.9890	0.9956	0.9956	**0.0390**	0.0456	0.0456	0.2479	0.2365	**0.2342**	0.0186	0.0161	**0.0159**
	80%	0.9055	0.9407	0.9363	**0.1055**	0.1407	0.1363	0.1917	0.1684	**0.1664**			
5	95%	0.9738	0.9976	0.9976	**0.0238**	0.0476	0.0476	0.2658	**0.2450**	0.2500	0.0195	**0.0164**	**0.0164**
	80%	0.8714	0.9048	0.9143	**0.0714**	0.1048	0.1143	0.2025	0.1694	**0.1688**			
6	95%	0.9818	1.0000	0.9974	**0.0318**	0.0500	0.0474	0.3129	0.2701	**0.2689**	0.0217	0.0174	**0.0173**
	80%	0.8805	0.9221	0.9143	**0.0805**	0.1221	0.1143	0.2140	0.1778	**0.1768**			
7	95%	0.9657	1.0000	1.0000	**0.0157**	0.0500	0.0500	0.3506	0.2867	**0.2824**	0.0236	**0.0181**	**0.0181**
	80%	0.9057	0.9429	0.9429	**0.1057**	0.1429	0.1429	0.2395	**0.1899**	0.1910			
8	95%	0.9714	1.0000	1.0000	**0.0214**	0.0500	0.0500	0.3016	0.3014	**0.2988**	0.0219	**0.0189**	**0.0189**
	80%	0.8508	0.9206	0.9206	**0.0508**	0.1206	0.1206	0.2216	**0.2007**	0.2014			
9	95%	0.9393	0.9929	0.9929	**0.0107**	0.0429	0.0429	0.4657	0.3230	**0.3214**	0.0274	**0.0203**	**0.0203**
	80%	0.8500	0.9107	0.9107	**0.0500**	0.1107	0.1107	0.2797	**0.2157**	**0.2157**			
10	95%	0.9633	0.9796	0.9755	**0.0133**	0.0296	0.0255	0.4215	0.2885	**0.2872**	0.0278	**0.0193**	**0.0193**
	80%	0.7755	0.8490	0.8612	**0.0245**	0.0490	0.0612	0.2777	**0.1980**	0.1991			

Figure 5.3 presents forecasts for all horizons $h = 1, \ldots, 10$ at the latest common test origin. Functional IC and CV are almost indistinguishable over most ages and are smoother than the raw target. Their largest errors occur at the oldest fertility ages, where the observed relative changes are most volatile because the rates in the denominator of (5.1) are small.

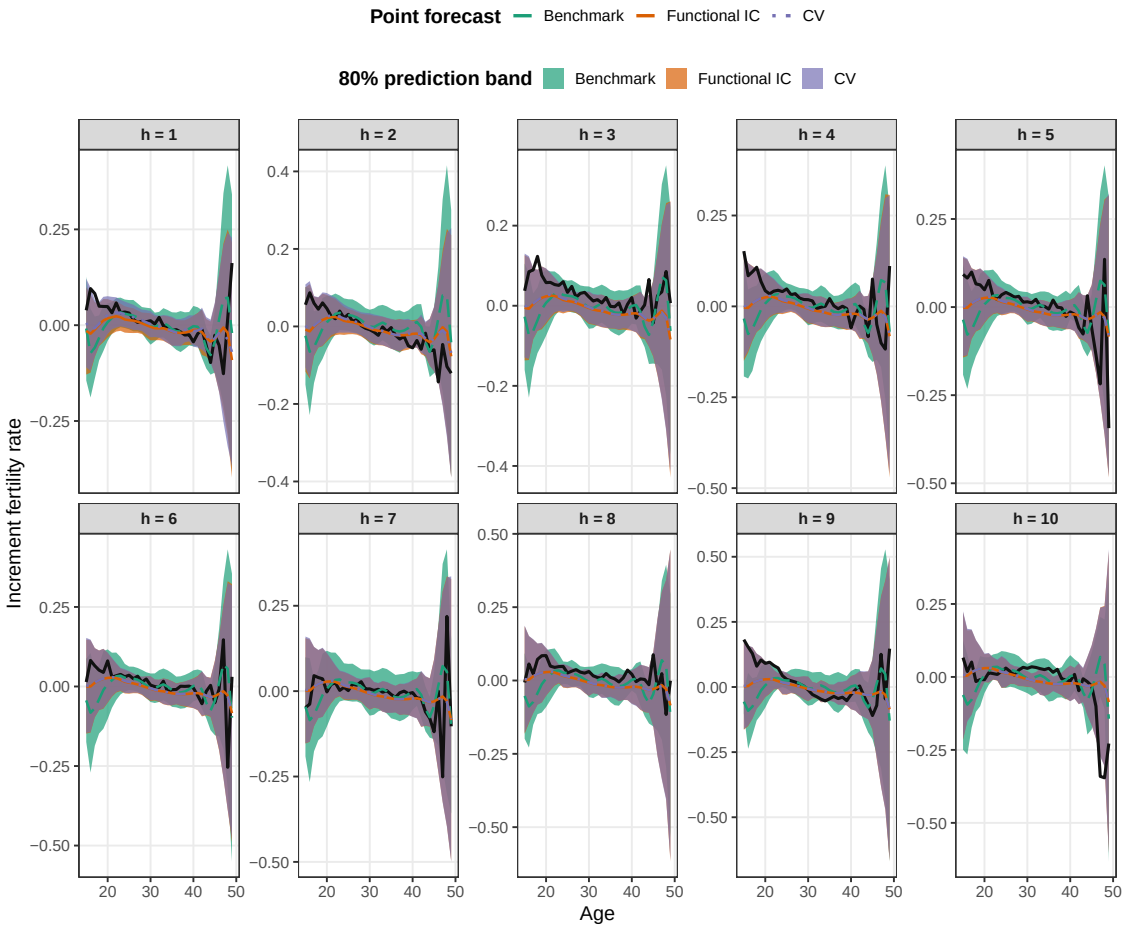


Figure 5.3: French fertility-rate increment forecasts for horizons $h = 1, \ldots, 10$ at the latest common test origin. Black solid lines are raw observed curves. Coloured lines are point forecasts, and each central 80% prediction band uses the colour of its corresponding method.

5.2.2 French Female and Male Mortality

Data source and preprocessing

The mortality data are annual age- and sex-specific death rates for the total population of France obtained from the Human Mortality Database (Human Mortality Database, 2026). The data, metadata, and download files are available from the Human Mortality Database France page. The database constructs death rates from death counts and the corresponding exposures to risk (Wilmoth et al., 2025). The sample covers 1861–2023 and contains 163 female and 163 male curves over ages 0–95.

Rates outside $(0, 1]$ are treated as missing and linearly interpolated over time within each age. The analysis uses base-10 log-mortality rates. The raw curves are smoothed using weighted penalised regression splines, with exposure-based weights and a monotonicity constraint above age 65 following Hyndman et al. (2007). Models are fitted to the smoothed curves, while RMSE, MAE, ECP, CPD, interval score, and WIS are evaluated against the raw log-mortality curves.

Figure 5.4 shows the familiar mortality profile: high infant mortality, a rapid decline during childhood, and an approximately monotone increase through adulthood and old age. The progression from early red curves to recent violet curves reflects the substantial secular decline in mortality. Smoothing removes local age-specific variation while retaining the long-run shape and the more pronounced young-adult mortality hump for males.

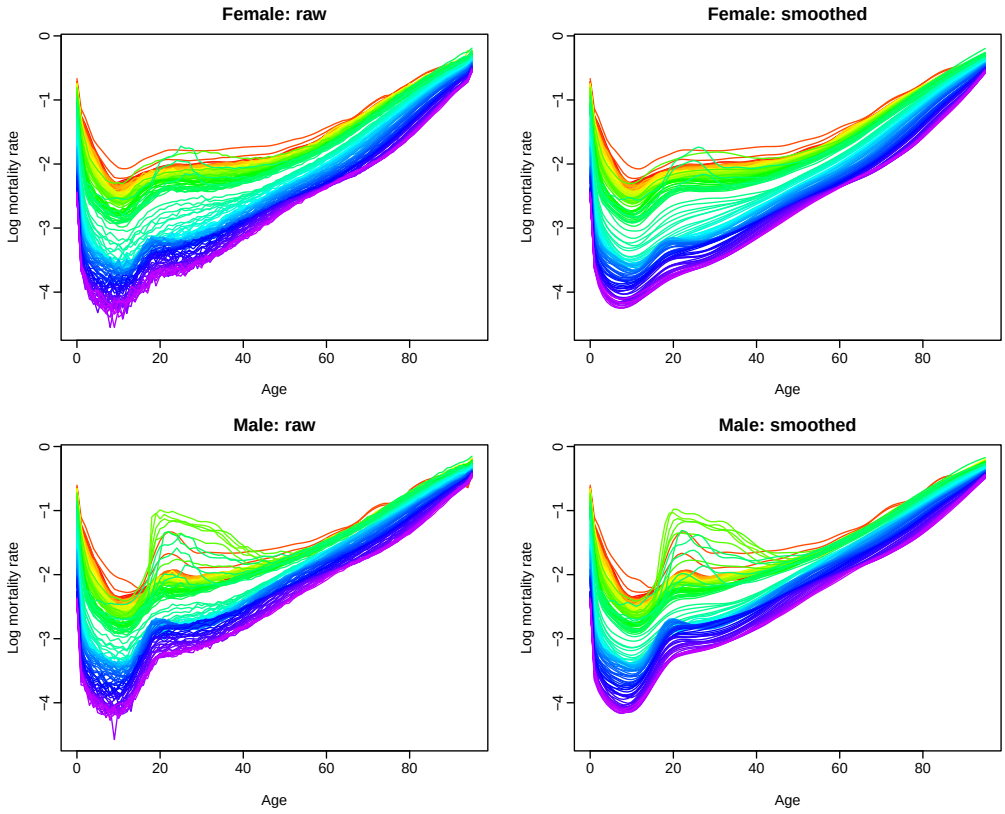
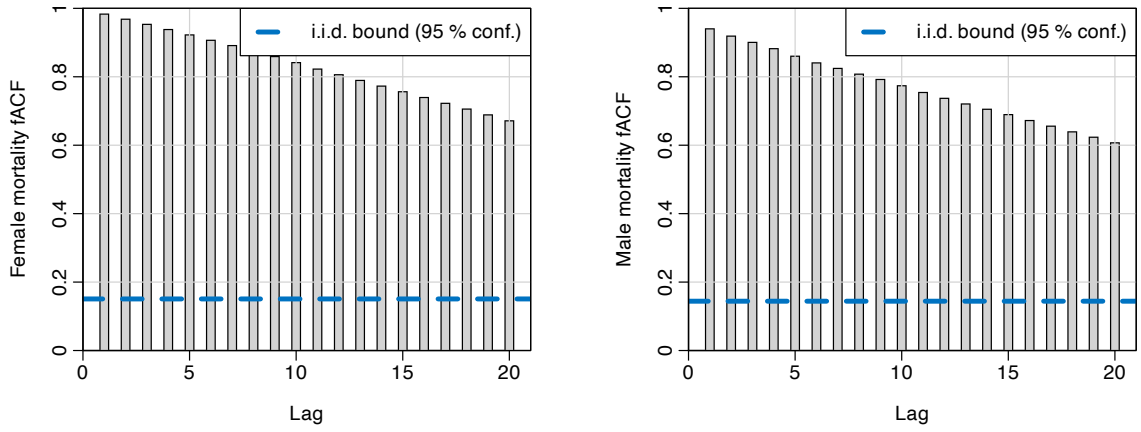


Figure 5.4: Rainbow plots of raw and smoothed French female and male log-mortality rates, 1861–2023. The left column contains raw curves and the right column smoothed curves.

Persistence and selected orders

Stationarity is rejected for female and male log mortality, with p-values of 0.0036 and 0.0040. The corresponding memory estimates are $\widehat{d} = 0.9250$ and 0.9909. Both lie in the nonstationary but mean-reverting range $1/2 < d < 1$, although the male estimate is very close to the unit-integration boundary. The fACFs in Figure 5.5 decay extremely slowly. The female value declines from 0.971 at lag one to 0.659 at lag 20, while the male value declines from 0.928 to 0.594. Every displayed fACF exceeds its white-noise bound.

Because the mortality estimates of d are already close to the benchmark value of one, changing d alone can provide only a modest adjustment. The more substantial model distinction is in the short-memory structure: Functional IC selects $(0, 1)$ for females and $(4, 0)$ for males, whereas CV selects $(2, 1)$ and $(2, 3)$. Mortality therefore provides the clearer assessment of whether selecting (p, q) improves forecasting when the memory order is already near one.



(a) Female log mortality

(b) Male log mortality

Figure 5.5: Functional autocorrelation functions of French female and male log mortality. The dashed blue line is the simulated upper 95% white-noise bound.

Forecasting results

Tables 5.3–5.6 show smaller improvements than in the fertility study. For females, Functional IC and CV reduce RMSE by approximately 5%–6% across the ten horizons, and their WIS advantage is clearest at medium and long horizons. For males, Functional IC gives the lowest RMSE and MAE at every horizon and generally gives the lowest WIS. CV becomes less accurate than the benchmark at longer horizons. The relatively complex CV-selected FARFIMA$(2, 0.9909, 3)$ specification therefore does not improve out-of-sample accuracy. This supports the adaptive penalty's intended role: retaining additional short-memory operators only when their reduction in residual dependence justifies the added complexity.

Table 5.3: Point forecast accuracy for French female log mortality over all ten horizons. The smallest RMSE is red and the smallest MAE is blue.

	RMSE			MAE		
h	Bench.	IC	CV	Bench.	IC	CV
1	0.0496	**0.0470**	**0.0470**	0.0317	**0.0297**	**0.0297**
2	0.0537	**0.0493**	0.0494	0.0352	**0.0318**	0.0319
3	0.0533	**0.0511**	0.0513	0.0356	**0.0340**	0.0342
4	0.0549	**0.0523**	0.0525	0.0375	**0.0362**	0.0363
5	0.0591	**0.0557**	0.0558	0.0416	**0.0396**	0.0397
6	0.0609	**0.0583**	0.0585	0.0440	**0.0427**	0.0430
7	0.0615	**0.0607**	0.0610	0.0450	**0.0449**	0.0452
8	0.0662	**0.0640**	0.0641	0.0488	**0.0479**	0.0481
9	0.0703	**0.0659**	0.0661	0.0524	**0.0504**	0.0506
10	0.0699	**0.0663**	0.0666	0.0530	**0.0514**	0.0517

Table 5.4: Point forecast accuracy for French male log mortality over all ten horizons. The smallest RMSE is red and the smallest MAE is blue.

	RMSE			MAE		
h	Bench.	IC	CV	Bench.	IC	CV
1	0.0385	**0.0369**	0.0373	0.0248	**0.0235**	0.0238
2	0.0409	**0.0402**	0.0411	0.0278	**0.0274**	0.0282
3	0.0439	**0.0434**	0.0448	0.0314	**0.0312**	0.0327
4	0.0476	**0.0468**	0.0488	0.0351	**0.0345**	0.0364
5	0.0530	**0.0513**	0.0535	0.0404	**0.0391**	0.0413
6	0.0582	**0.0567**	0.0592	0.0455	**0.0445**	0.0469
7	0.0628	**0.0613**	0.0640	0.0494	**0.0484**	0.0510
8	0.0671	**0.0655**	0.0685	0.0535	**0.0524**	0.0552
9	0.0725	**0.0706**	0.0735	0.0584	**0.0570**	0.0598
10	0.0765	**0.0748**	0.0780	0.0619	**0.0605**	0.0635

Raw-scale interval evaluation reveals substantial short-horizon under-coverage. At $h = 1$, 95% ECP ranges from 0.732 to 0.775 for females and from 0.762 to 0.774 for males. Coverage generally improves with horizon. At $h = 10$, female 95% ECP is 0.938 for the benchmark, 0.959 for Functional IC, and 0.960 for CV; the corresponding male values are 0.928, 0.931, and 0.929. Functional IC generally gives the lowest mortality interval scores, especially for males. The under-coverage is consistent with calibrating intervals from smoothed validation residuals and evaluating them against noisier raw observations.

Table 5.5: Interval forecast performance for French female log mortality over all ten horizons. The smallest CPD is red, the smallest interval score is blue, and the smallest WIS is purple. WIS combines both coverage levels and is reported once per horizon in the 95% row.

		ECP			CPD			Score			WIS		
h	CP	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV
1	95%	0.7754	0.7342	0.7317	**0.1746**	0.2158	0.2183	**0.3478**	0.3825	0.3832	**0.0167**	**0.0167**	0.0168
	80%	0.5908	0.5288	0.5308	**0.2092**	0.2712	0.2692	**0.1717**	0.1739	0.1745			
2	95%	0.8177	0.7856	0.7869	**0.1323**	0.1644	0.1631	**0.3362**	0.3528	0.3528	0.0178	**0.0169**	**0.0169**
	80%	0.6094	0.6185	0.6089	0.1906	**0.1815**	0.1911	0.1853	**0.1740**	0.1757			
3	95%	0.8483	0.8460	0.8456	**0.1017**	0.1040	0.1044	**0.2960**	0.3086	0.3120	0.0171	**0.0167**	0.0168
	80%	0.6834	0.6934	0.6898	0.1166	**0.1066**	0.1102	0.1749	**0.1706**	0.1719			
4	95%	0.8627	0.8655	0.8613	0.0873	**0.0845**	0.0887	0.3028	**0.2986**	0.3045	0.0179	**0.0172**	0.0174
	80%	0.6918	0.7150	0.7060	0.1082	**0.0850**	0.0940	0.1841	**0.1758**	0.1771			
5	95%	0.8929	0.9038	0.9072	0.0571	0.0462	**0.0428**	0.3032	0.2802	**0.2796**	0.0192	**0.0181**	**0.0181**
	80%	0.7054	0.7396	0.7381	0.0946	**0.0604**	0.0619	0.1967	**0.1852**	**0.1852**			
6	95%	0.8880	0.9115	0.9089	0.0620	**0.0385**	0.0411	0.3114	**0.2981**	0.2995	0.0201	**0.0193**	0.0194
	80%	0.7286	0.7484	0.7443	0.0714	**0.0516**	0.0557	0.2049	**0.1946**	0.1955			
7	95%	0.9117	0.9315	0.9282	0.0383	**0.0185**	0.0218	0.3047	**0.2986**	0.3006	0.0203	**0.0201**	0.0202
	80%	0.7681	0.7906	0.7922	0.0319	0.0094	**0.0078**	0.2064	**0.2039**	0.2047			
8	95%	0.9219	0.9346	0.9317	0.0281	**0.0154**	0.0183	0.3179	**0.3027**	0.3029	0.0220	**0.0212**	**0.0212**
	80%	0.7818	0.8015	0.8027	0.0182	**0.0015**	0.0027	0.2255	0.2148	**0.2147**			
9	95%	0.9326	0.9498	0.9485	0.0174	**0.0002**	0.0015	0.3304	**0.2990**	**0.2990**	0.0233	**0.0219**	**0.0219**
	80%	0.7953	0.8505	0.8480	**0.0047**	0.0505	0.0480	0.2378	0.2202	**0.2198**			
10	95%	0.9382	0.9590	0.9603	0.0118	**0.0090**	0.0103	0.3527	0.3249	**0.3248**	0.0242	**0.0235**	**0.0235**
	80%	0.8177	0.8841	0.8835	**0.0177**	0.0841	0.0835	0.2512	0.2485	**0.2476**			

Table 5.6: Interval forecast performance for French male log mortality over all ten horizons. The smallest CPD is red, the smallest interval score is blue, and the smallest WIS is purple. WIS combines both coverage levels and is reported once per horizon in the 95% row.

		ECP			CPD			Score			WIS		
h	CP	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV
1	95%	0.7738	0.7721	0.7621	**0.1762**	0.1779	0.1879	0.2631	**0.2608**	0.2687	0.0130	**0.0124**	0.0127
	80%	0.5908	0.6033	0.5983	0.2092	**0.1967**	0.2017	0.1347	**0.1276**	0.1301			
2	95%	0.8242	0.8329	0.8142	0.1258	**0.1171**	0.1358	0.2420	**0.2359**	0.2495	0.0137	**0.0135**	0.0140
	80%	0.6393	0.6224	0.5977	**0.1607**	0.1776	0.2023	0.1421	**0.1415**	0.1474			
3	95%	0.8537	0.8338	0.8234	**0.0963**	0.1162	0.1266	**0.2484**	0.2504	0.2626	**0.0150**	**0.0150**	0.0157
	80%	0.6916	0.6716	0.6372	**0.1084**	0.1284	0.1628	**0.1568**	0.1572	0.1634			
4	95%	0.8797	0.8897	0.8698	0.0703	**0.0603**	0.0802	0.2555	**0.2512**	0.2639	0.0164	**0.0161**	0.0169
	80%	0.7206	0.7197	0.6899	**0.0794**	0.0803	0.1101	0.1709	**0.1682**	0.1743			
5	95%	0.9028	0.8958	0.8805	**0.0472**	0.0542	0.0695	0.2798	**0.2754**	0.2839	0.0187	**0.0182**	0.0189
	80%	0.7197	0.7351	0.7049	0.0803	**0.0649**	0.0951	0.1950	**0.1895**	0.1939			
6	95%	0.9151	0.9109	0.8990	**0.0349**	0.0391	0.0510	0.3130	**0.3042**	0.3116	0.0208	**0.0203**	0.0211
	80%	0.7245	0.7245	0.6974	**0.0755**	**0.0755**	0.1026	0.2147	**0.2093**	0.2141			
7	95%	0.9254	0.9276	0.9167	0.0246	**0.0224**	0.0333	0.3232	**0.3142**	0.3196	0.0222	**0.0217**	0.0225
	80%	0.7577	0.7719	0.7319	0.0423	**0.0281**	0.0681	0.2264	**0.2215**	0.2268			
8	95%	0.9277	0.9265	0.9149	**0.0223**	0.0235	0.0351	0.3491	**0.3378**	0.3473	0.0240	**0.0234**	0.0243
	80%	0.7755	0.7784	0.7517	0.0245	**0.0216**	0.0483	0.2460	**0.2389**	0.2454			
9	95%	0.9277	0.9357	0.9252	0.0223	**0.0143**	0.0248	0.3780	**0.3634**	0.3724	0.0260	**0.0253**	0.0262
	80%	0.7800	0.7800	0.7567	**0.0200**	**0.0200**	0.0433	0.2644	**0.2557**	0.2643			

Table 5.6– continued

		ECP			CPD			Score			WIS		
h	CP	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV	Bench.	IC	CV
10	95%	0.9277	0.9310	0.9290	0.0223	**0.0190**	0.0210	0.4281	**0.4037**	0.4183	0.0285	**0.0275**	0.0286
	80%	0.7975	0.7917	0.7695	**0.0025**	0.0083	0.0305	0.2963	**0.2832**	0.2935			

Figures 5.6 and 5.7 give the forecasts for all horizons $h = 1, \ldots, 10$ at the latest common test origin. The three forecasts are close because d is near one and the broad mortality profile dominates local differences. Nevertheless, the accumulated results in Tables 5.3 and 5.4 show that Functional IC provides the most reliable short-memory order selection, particularly for male mortality.

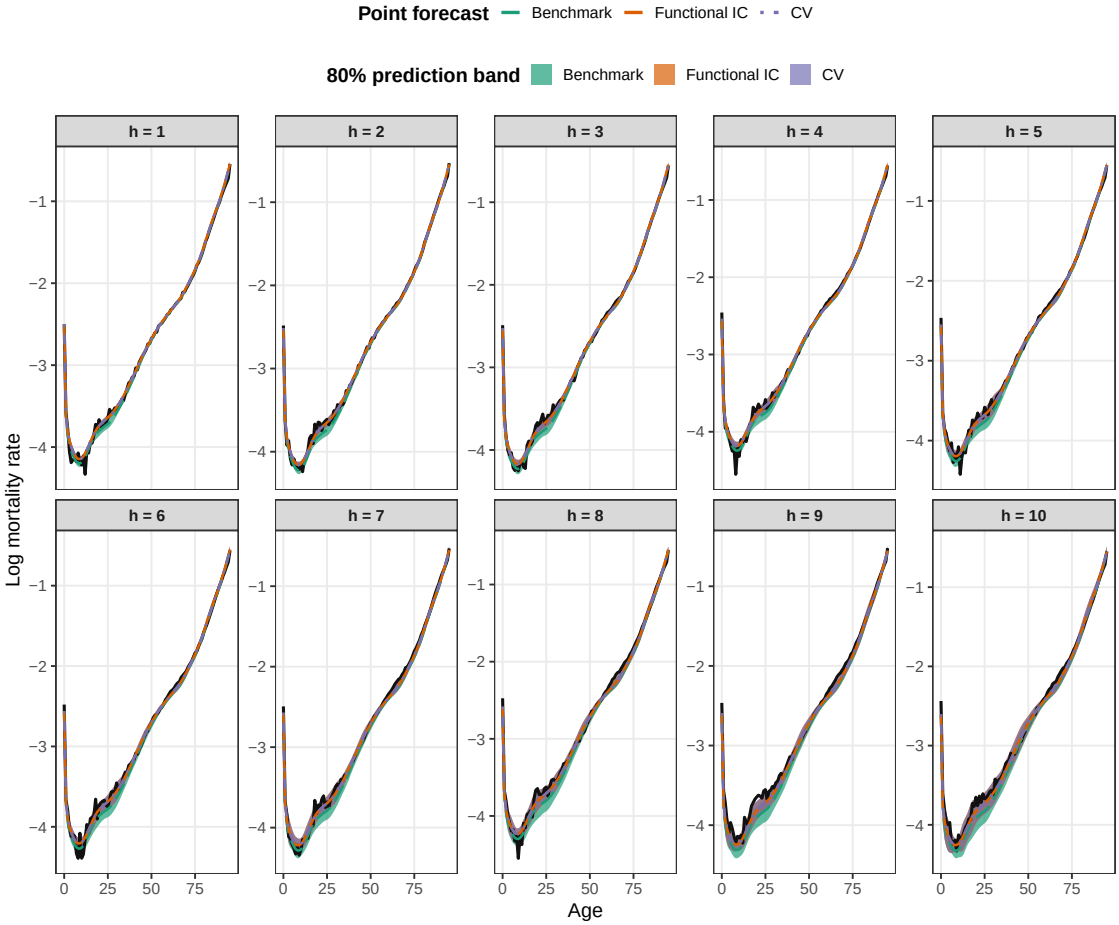


Figure 5.6: French female log-mortality forecasts for horizons $h = 1, \ldots, 10$ at the latest common test origin. Black solid lines are raw observed curves. Coloured lines are point forecasts, and each central 80% prediction band uses the colour of its corresponding method.

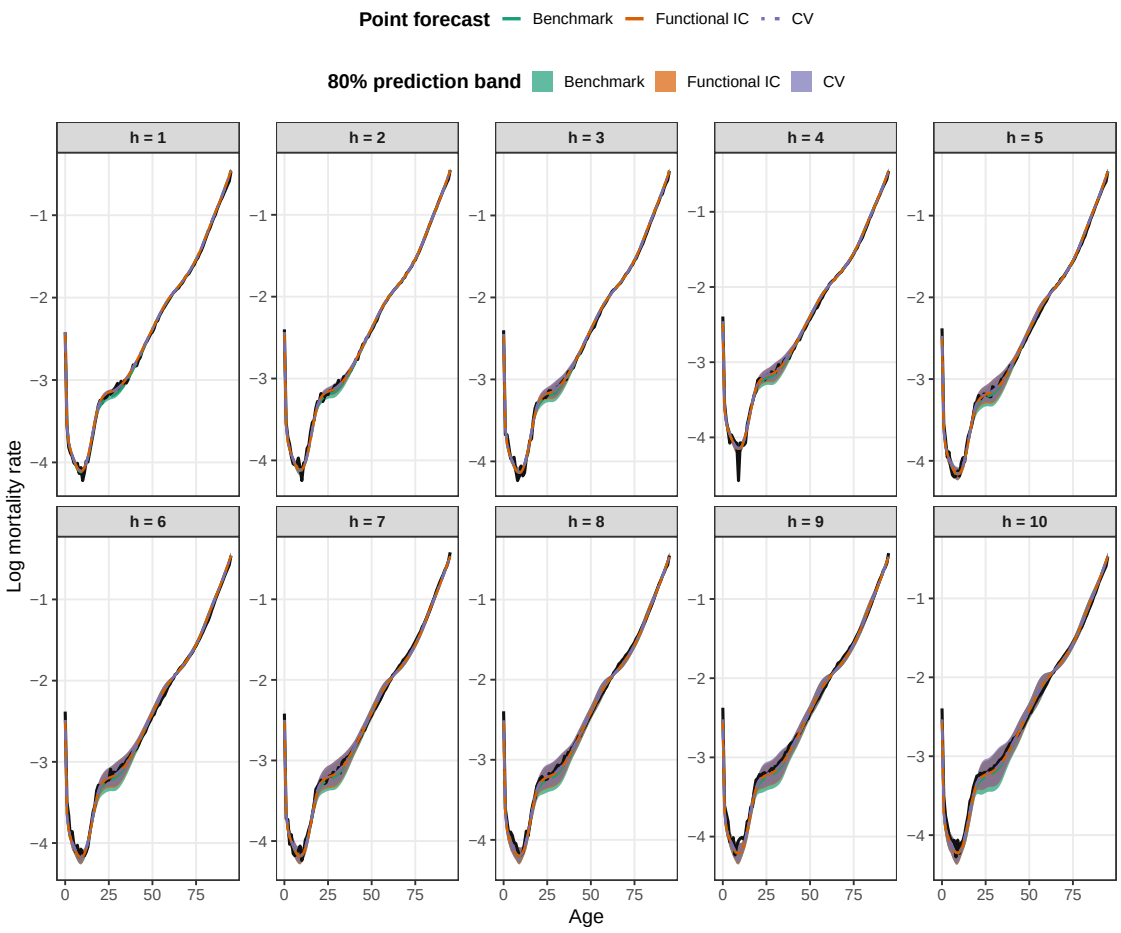


Figure 5.7: French male log-mortality forecasts for horizons $h = 1, \ldots, 10$ at the latest common test origin. Black solid lines are raw observed curves. Coloured lines are point forecasts, and each central 80% prediction band uses the colour of its corresponding method.

5.3 Comparison and Discussion

The two empirical studies illustrate complementary benefits of the proposed framework. For fertility-rate changes, $\widehat{d} = 0.3886$ is far from the benchmark value $d = 1$, while the Functional IC and CV short-memory orders are similar. The large and nearly identical forecast gains from the two data-driven methods therefore point mainly to correct estimation of fractional integration. For mortality, $\widehat{d}$ is already close to one, so the smaller differences in performance arise mainly from selecting (p, q). Functional IC improves both female and male forecasts without adopting the unnecessarily rich male order selected by CV.

This decomposition should be understood as an evidence-based interpretation rather than a formal causal attribution. A direct robustness check could compare models that vary only d while fixing (p, q) and models that vary only (p, q) while fixing d. Even without that additional experiment, the contrast is informative: estimating d matters most when the unit-integration benchmark is incompatible with the data, whereas adaptive selection of (p, q) matters more when d is already near one.

Two further qualifications are important. First, the fACF and fractional estimate provide evidence of persistence but are not a stand-alone test of the complete FARFIMA specification. Second, the intervals are calibrated from smoothed curves but assessed against raw observations. This is appropriate for evaluating observed demographic rates, but introduces measurement variation absent from the calibration target and is most visible in short-horizon mortality coverage. A useful sensitivity analysis would recalibrate nonconformity scores to raw validation curves or explicitly include an observation-error component.

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